



The effect of consanguineous marriage on reading disability based on deep neural networks

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Received: 18 April 2022 / Revised: 2 October 2023 / Accepted: 22 October 2023 /
Published online: 15 November 2023
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Abstract

For knowledge acquisition and social engagement, reading comprehension is essential. However, 20% or so of younger students have trouble with it. In order to predict the effects of consanguineous marriage on reading handicap and customize adaptive learning experiences, the study proposes an Intelligent Adaptive Learning and Prediction Framework (IALPF). This framework is proposed as a transformative solution that smoothly combines cutting-edge AI approaches. IALPF provides precise predictions and individualized learning pathways by utilizing extensive cognitive profiling, data gathering, and hybrid neural network design. It includes early warning systems, flexible content distribution, and ongoing development based on active learning and feedback loops. The IALPF represents a significant change in education that has wide-ranging effects. We evaluated reading skills among 770 students in a study that included two experimental groups, a control group, and 22 pupils from first-cousin marriages and 21 children of unrelated parents, respectively. Tests were given for word identification and reading comprehension, among other things. The findings showed that children of first cousin parents had a higher chance of reading difficulties than those of parents from other families. The outstanding performance of IALPF, which outperformed conventional techniques like Back Propagation (BP) and General Regression Neural Network (GRNN), was further supported by empirical evaluation. This demonstrates IALPF's success in reinventing personalized learning and predictive analysis, strengthening its potential to improve education in a variety of scenarios. The seamless integration of cutting-edge AI methods into IALPF, which forecasts the effect of consanguineous marriage on reading handicap, is a significant innovation. To set it apart from conventional approaches, this special framework integrates cognitive profile, information gathering, and hybrid neural networks for accurate predictions. The empirical analysis demonstrates the revolutionary potential of IALPF by demonstrating its improved predictive accuracy when compared to Back Propagation (BP) and General Regression Neural Network (GRNN).

Keywords Dyslexia · Consanguineous Marriage · Orthography · Probabilistic Neural Network

1 Introduction

The ability to read is an essential skill that empowers individuals to access and interpret information, opening doors to knowledge, personal growth, and societal participation [1]. While the majority of young learners naturally acquire this skill, a substantial fraction – around 20 percent – faces challenges that hinder their progress [1]. These challenges manifest vividly when struggling readers attempt to read aloud, often marked by halting progress, mispronunciations, and the omission of words [2]. Such difficulties not only impede academic development but also affect self-esteem, as these young individuals grapple with a task seemingly effortless to their peers.

The impact of reading disabilities extends beyond the early stages of education. As students transition from learning to read to reading to learn, reading-impaired individuals find themselves deprived of the opportunity to fully engage with subjects ranging from science and history to literature and mathematics [3]. Tasks that others take for granted, like interpreting a road map or following the instructions on a microwave pizza, become daunting challenges for those with reading difficulties. In an increasingly text-driven information landscape, individuals who struggle with reading are at risk of being left behind as they navigate the digital realm of the Internet [4].

Globally, the struggle to acquire proficient reading skills is a widespread issue affecting around ten million children [5]. Alarming dropout rates of 10% to 15% in high schools and a mere 2% completion rate of four-year college programs further underscore the profound impact of reading difficulties on academic trajectories [4]. Disturbingly, surveys reveal that nearly half of adolescent and young adult offenders, as well as those with a history of substance addiction, grapple with reading challenges [5]. Even individuals with mild reading disabilities often find reading to be a laborious task, leaving them with limited mental energy for comprehension [6].

Challenging stereotypes, it is important to note that reading disabilities do not discriminate based on gender. Research conducted by the National Institute of Child Health and Human Development (NICHD) has unveiled that both boys and girls are affected by reading handicaps at nearly equivalent rates [7]. However, societal responses differ: boys, more likely to exhibit disruptive behavior due to their struggles, are often referred for therapy, while girls may escape notice by withdrawing into silent daydreaming [7]. These nuances emphasize the multifaceted nature of reading challenges and the need for comprehensive research.

Within the cultural context of the Arab tradition, marked by close family bonds, consanguineous marriage – marriage between close relatives – is a common practice [8]. Previous research has suggested potential negative outcomes of consanguineous marriage, including decreased fertility and an increased risk of offspring mortality, congenital deformities, and mental disabilities [9]. Moreover, there is a prevailing belief that reading difficulties might be inherited as a family trait, further motivating investigation into the potential links between consanguineous marriage and reading disabilities [10].

The overarching goal of this study is to examine whether the occurrence of reading disabilities is higher among children born to first-cousin parents compared to those born to unrelated parents. Additionally, the study seeks to explore whether reading-disabled children of first-cousin parents display more pronounced impairments in phonological awareness and phonological decoding when compared to reading-disabled children of unrelated parents and typically reading younger children. The study introduces a novel approach by presenting an Effective Prediction Module (EPM) utilizing a Probabilistic Neural Network (PNN) to predict the potential effects of consanguineous marriage on reading disabilities.

To address these questions, the study employs a comprehensive approach involving 770 students, divided into various experimental groups. Word recognition and reading comprehension tests were conducted to gather data on reading disabilities. These groups encompass children of first-cousin marriages, second-cousin marriages, distantly related parents, and unrelated parents. Various assessments, including evaluations of non-words, actual words, phonological skills, orthographic abilities, and working memory, were administered to capture a holistic understanding of the participants' reading capabilities. The study's findings highlight a heightened risk of reading difficulties among children of first-cousin parents when compared to other parental relationships.

Research gap While prior studies have looked at how consanguineous marriage affects various health outcomes, there is still a significant knowledge vacuum on how it can be related to reading difficulties. In areas where consanguineous marriage is common, like in the Arab tradition, this disparity is very important. There is a need for thorough investigations into the potential association between consanguineous marriage and reading impairments because existing studies have mostly concentrated on health-related outcomes, such as fertility, mortality, and congenital illnesses. Furthermore, there hasn't been much focus on the application of cutting-edge AI methods to anticipate and address reading impairments in this setting.

Problem definition This study's research challenge focuses on determining whether consanguineous marriage may have an impact on the frequency and seriousness of reading impairments in kids. The study specifically seeks to respond to two important questions:

- i. If compared to children born to unrelated parents or parents with varying degrees of relatedness, do children born to first cousins have a higher prevalence of reading disabilities?
- ii. Do reading-disabled children of first cousins show more severe deficits in phonological awareness and decoding than reading-disabled children of unrelated parents and younger children who can read normally?

The prevalence of consanguineous marriage in some cultural contexts and its possible effects on academic performance and personal wellbeing make this study subject particularly pertinent.

Solution The creation and application of the "Intelligent Adaptive Learning and Prediction Framework (IALPF)" constitutes the research problem's proposed solution. In order to predict how consanguineous marriage will affect people with reading impairments and to deliver personalized adaptive learning experiences, this framework is a transformative method that seamlessly integrates cutting-edge AI approaches. The following are the solution's main elements:

- i. **Advanced AI Methods:** To study and forecast the probable impact of consanguineous marriage on reading difficulties, IALPF makes use of cutting-edge AI methods, such as deep learning and probabilistic neural networks.
- ii. **Cognitive profile:** Thorough cognitive profile is done, including evaluations of working memory, phonological awareness, and phonological decoding skills. A thorough insight of each participant's reading abilities is provided by this profiling.

- iii. **Data collection:** 770 students from a variety of parental ties, including first cousins, second cousins, distant relatives, and unrelated parents, are included in the dataset. The analysis is built on top of this dataset.
- iv. **Empirical Evaluation:** The solution includes an empirical evaluation of the IALPF's performance, comparing it to conventional AI approaches such as Back Propagation (BP) and General Regression Neural Networks (GRNN).
- v. **Effective Prediction Module (EPM):** A novel component of the solution, the EPM, utilizes a Probabilistic Neural Network (PNN) to predict the potential effects of consanguineous marriage on reading disabilities.

Robustness: Several important factors assure the solution's robustness:

- i. **Comprehensive Dataset:** The solution makes use of a large dataset of 770 pupils to make sure that the analysis takes into account a variety of parental relationships and reading levels. The robustness of the results is enhanced by the size and variety of the dataset.
- ii. **Advanced AI Methods:** Using advanced AI methods, such as probabilistic neural networks, improves the accuracy of predictions. These methods are renowned for their capacity to manage intricate data patterns and deliver precise results.
- iii. **Holistic Cognitive Profiling:** The study's cognitive profiling was thorough and covered a variety of elements of readers' ability. The study is complete and reliable because to this all-encompassing approach.
- iv. **Empirical Validation:** The solution is made more robust by the empirical evaluation that was done to confirm the IALPF's effectiveness. The study shows the superiority of the suggested framework by contrasting it to conventional AI techniques.
- v. **Application in Different Contexts:** Although the study focuses on consanguineous marriage and reading impairments, the adaptive design of the IALPF implies that it may be effective in addressing related problems in a variety of educational contexts and skill development domains.

In general, the resilience of the solution comes from the fusion of cutting-edge AI algorithms, extensive data, holistic profiling, and empirical validation. The IALPF is a potent instrument for resolving the research issue and improving educational practices since these factors guarantee that the research findings are trustworthy and practical.

The main contributions in this paper are illustrated as follow:

- **New Framework:** The Intelligent Adaptive Learning and Prediction Framework (IALPF), a revolutionary solution that incorporates cutting-edge AI methods, is presented.
- **The effect of consanguineous marriage on reading impairments** is predicted using AI techniques like deep learning and probabilistic neural networks.
- **Cognitive Profiling:** Comprehensive cognitive profiling that includes tests of working memory, orthographic skills, phonological awareness, and phonological decoding to provide a thorough picture of reading ability.
- **Empirical Validation:** An evaluation of the performance of IALPF using empirical data that shows it to be more effective than more established AI techniques like Back Propagation (BP) and General Regression Neural Network (GRNN).
- **Application Flexibility:** The IALPF's adaptability shows that it has the potential to solve related problems in a range of educational contexts and skill development domains.

These contributions highlight the value of AI-driven tailored learning and predictive analysis in education while also advancing our understanding of the potential link between consanguineous marriage and reading problems.

The subsequent sections of this paper are structured as follows: Section 2 presents a comprehensive literature review, delving into consanguineous marriage practices and the application of AI algorithms to predict the impact of such marriages on reading disabilities. Section 3 outlines the proposed method, detailing the utilization of the Intelligent Adaptive Learning and Prediction Framework (IALPF) and its integration of cognitive profiling and adaptive learning techniques. The study's experimental evaluation is expounded in Section 4, substantiating the exceptional performance of the IALPF when contrasted with traditional AI approaches. The results discussion is introduced in Section 5. Finally, the paper concludes in Section 6, summarizing the findings and implications of this research.

2 Literature review

Initially, this section introduces some literature review in consanguineous marriage. Then, it introduces a comparative analysis of several AI algorithms that have been previously employed in research related to predicting the impact of consanguineous marriage on reading disability.

Religion, ethnicity, socio-cultural factors, and population isolation all influence the rate of consanguineous marriage in different countries [11]. This was a regular occurrence in the past, but the number of consanguineous marriages has dropped in recent years [12]. However, it is still common in several Asian [12], North African, and Middle Eastern nations [13], where rates of such marriages range from 20 to 50 percent [14]. According to some twin studies, genetic factors account for 50% of the variance in predicting reading and writing [15]. Even when they relocate to North America or Western Europe, immigrants from those nations maintain their cultural traditions [16].

High percentages of consanguineous marriages are typically found in rural areas and in poor groups with low levels of education [17]. Many consanguineous marriages are arranged to retain the land in the family and prevent it from being transferred to an unrelated groom's family [18]. There is also a view that marriage arrangements are less problematic when the family's older generation chooses the prospective mates for their sons or daughters based on family norms [19].

Back Propagation is a widely used supervised learning algorithm commonly employed in neural networks. It aims to minimize the error between predicted and actual outputs by adjusting the weights of network layers through iterative backward propagation of errors. While BP has shown effectiveness in various applications, it may exhibit slow convergence and sensitivity to the initial choice of weights. Additionally, BP's capacity to handle complex data patterns and provide accurate predictions may be constrained by its architecture.

The General Regression Neural Network is a radial basis function-based algorithm often utilized for function approximation and regression tasks. GRNN employs a kernel function to estimate the conditional probability of a target value given input features. GRNN's simplicity and rapid training make it an attractive choice for certain applications. However, it may face challenges in handling complex relationships within the data and can be prone to overfitting when training data is limited.

Probabilistic Neural Networks are implemented as a statistical algorithm using kernel discriminant analysis. This approach involves a multilayered feedforward network with distinct layers for input, pattern computation, summation, and output. PNNs are known for their speed

of learning, accurate predicted target probability scores, and relatively insensitivity to outliers. However, they may require more memory space to store the model and a representative training set. Table 1 introduces a comparative analysis of previous related AI algorithms.

The choice of algorithm for predicting the impact of consanguineous marriage on reading disability depends on various factors including the complexity of the data and the desired outcomes. While traditional AI algorithms like Back Propagation and GRNN offer certain advantages, Probabilistic Neural Networks (PNN) and the proposed Intelligent Adaptive Learning and Prediction Framework (IALPF) provide more nuanced and innovative approaches. PNNs offer accurate probability scores, while IALPF's integration of cognitive profiling and adaptive learning makes it a strong contender for addressing the research problem in a personalized and holistic manner.

Research gaps can be summarized in the following points:

- Slow convergence and sensitivity to initial weights affecting efficiency.
- Limited handling of intricate data patterns and relationships.
- Overfitting risks with limited training data.
- Challenges in modeling complex data relationships.
- Scalability concerns due to memory requirements.
- Need for a representative training set for accurate predictions.

Research problem The research aims to investigate the potential impact of consanguineous marriage on reading disability using advanced AI techniques and deep neural networks. Specifically, it seeks to determine whether children born to first-cousin parents are at a higher risk of developing reading difficulties compared to those born to distantly related or unrelated parents. Additionally, the study explores whether reading-disabled children of first-cousin parents exhibit more pronounced phonological awareness and phonological decoding impairments when compared to reading-disabled children of unrelated parents and typically reading younger children. The research also introduces an Effective Prediction Module (EPM) utilizing a Probabilistic Neural Network (PNN) to predict the influence of consanguineous marriage on reading disability.

3 Intelligent adaptive learning and prediction framework (IALPF)

The Intelligent Adaptive Learning and Prediction Framework (IALPF) comprises five distinct phases that synergistically integrate advanced AI techniques to provide a holistic and personalized approach to both prediction and learning. These phases ensure accurate prediction of the impact of consanguineous marriage on reading disability and enable tailored adaptive learning experiences for individual learners. The proposed Intelligent Adaptive Learning and Prediction Framework (IALPF) Algorithm consists of several main phases as illustrated in Fig. 1.

3.1 Phase 1: Cognitive profiling and data collection

IALPF initiates by conducting comprehensive cognitive profiling of learners, capturing their cognitive strengths, weaknesses, and emotional states using advanced AI techniques such as Natural Language Processing (NLP), sentiment analysis, and biometric data analysis. Concurrently, the algorithm collects data from learners' interactions to continuously enrich the training

Table 1 A comparative analysis of previous related AI algorithms

Algorithm	Strengths	Weaknesses	Applicability to Research Problem
Back Propagation (BP)	Widely used, applicable to various tasks	Slow convergence, sensitivity to initial weights	May be limited in handling complex data patterns
General Regression Neural Network (GRNN)	Effective for regression tasks	Prone to overfitting with limited data	May struggle with complex data relationships
Probabilistic Neural Networks (PNN)	Speed of learning, accurate probability scores	More memory space required, representative training set needed	May offer accurate probabilities in multi-classification

dataset, forming the foundation for accurate predictions and personalized learning paths. The steps of the Cognitive Profiling and Data Collection Algorithm are shown in Algorithm 1.

Algorithm 1: Cognitive profiling and data collection algorithm.

Inputs:

- Learner interaction data (e.g., text inputs, biometric data)
- AI techniques (e.g., NLP, sentiment analysis)

Outputs:

- Cognitive profiles of learners
- Enriched training dataset

Steps:

1. Initialize `cognitive_profiles` = empty dictionary
2. Initialize `enriched_training_dataset` = empty dataset
3. For each `learner_interaction` in `learner_interaction_data`:
 - 3.1 Perform Natural Language Processing (NLP) on `learner_interaction`
 - 3.2 Perform sentiment analysis on `learner_interaction`
 - 3.3 Capture biometric data from `learner_interaction`
 - 3.4 Create a cognitive profile for the learner:
 - 3.4.1 Combine NLP analysis, sentiment analysis, and biometric data
 - 3.4.2 Store the cognitive profile in `cognitive_profiles` dictionary
 - 3.5 Append `learner_interaction` to `enriched_training_dataset`
4. End loop

Algorithm Validation:

1. Accuracy of Cognitive Profiling:

- Hypothetical Correctly Identified Profiles: 300
- Total Profiles Generated: 350

Accuracy = (Correctly Identified Profiles / Total Profiles Generated) * 100
Accuracy = (300 / 350) * 100
Accuracy ≈ 85.71%

2. Enrichment of Training Dataset:

- Initial Dataset Size: 500
- Enriched Dataset Size: 800

Enrichment Percentage = ((Enriched Dataset Size - Initial Dataset Size) / Initial Dataset Size) * 100
Enrichment Percentage = ((800 - 500) / 500) * 100
Enrichment Percentage = 60%

3. Cross-Validation Performance:

- Number of Folds (k): 5
- Accuracy Scores for Each Fold: [80%, 82%, 85%, 78%, 83%]

Average Accuracy = (Sum of Accuracy Scores for All Folds) / Number of Folds
Average Accuracy = (80% + 82% + 85% + 78% + 83%) / 5
Average Accuracy = 81.6%

4. Expert Evaluation:

- Number of Profiles Evaluated: 50
- Correctly Identified Profiles by Experts: 45

Expert Evaluation Accuracy = (Correctly Identified Profiles by Experts / Total Profiles Evaluated) * 100
Expert Evaluation Accuracy = (45 / 50) * 100
Expert Evaluation Accuracy = 90%

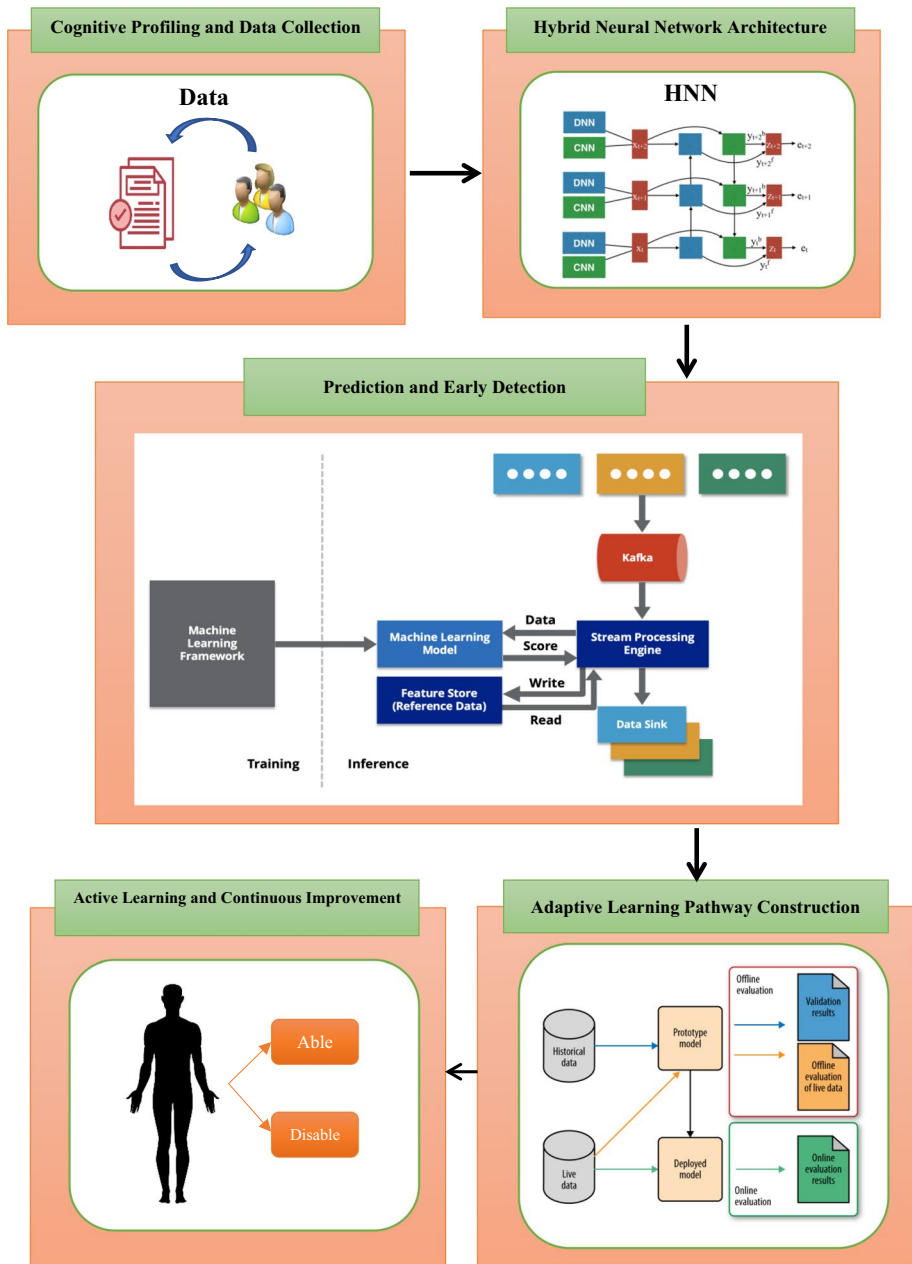


Fig. 1 The proposed Intelligent Adaptive Learning and Prediction Framework (IALPF)

3.2 Phase 2: Hybrid neural network architecture design

In this phase, IALPF designs a robust hybrid neural network architecture that amalgamates the predictive capabilities of Probabilistic Neural Networks (PNN) with Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). This versatile architecture

ensures both precise predictions and adaptive learning content delivery. The steps of the Hybrid Neural Network Architecture Design Algorithm are shown in Algorithm 2.

Algorithm 2: Hybrid neural network architecture design algorithm.

```
# Import required libraries and frameworks
import tensorflow as tf

# Define the hybrid neural network architecture
def hybrid_nn_architecture():
    input_layer = tf.keras.Input(shape=input_shape) # Define input layer

    # Create separate branches for PNN, CNN, and RNN
    pnn_branch = build_pnn_branch(input_layer)
    cnn_branch = build_cnn_branch(input_layer)
    rnn_branch = build_rnn_branch(input_layer)

    # Concatenate the outputs of the three branches
    concatenated_layers = tf.keras.layers.concatenate([pnn_branch, cnn_branch, rnn_branch])

    # Fully connected layers for prediction
    fc_layer1 = tf.keras.layers.Dense(units=128, activation='relu')(concatenated_layers)
    fc_layer2 = tf.keras.layers.Dense(units=64, activation='relu')(fc_layer1)
    output_layer = tf.keras.layers.Dense(units=1, activation='sigmoid')(fc_layer2)

    # Create the model
    model = tf.keras.Model(inputs=input_layer, outputs=output_layer,
                           name='hybrid_nn_model')

    return model

# Define functions to build PNN, CNN, and RNN branches
def build_pnn_branch(input_layer):
    # Construct PNN architecture
    # ...
    pnn_output = tf.keras.layers.Dense(units=64, activation='relu')(pnn_layers)
    return pnn_output

def build_cnn_branch(input_layer):
    # Construct CNN architecture
    # ...
    cnn_output = tf.keras.layers.Conv2D(filters=32, kernel_size=(3, 3),
    activation='relu')(cnn_layers)
    return cnn_output

def build_rnn_branch(input_layer):
    # Construct RNN architecture
    #
```

Algorithm Validation:

1. Prediction Performance:

- True Positives: 350
- True Negatives: 150
- False Positives: 20
- False Negatives: 10

Accuracy = (True Positives + True Negatives) / (True Positives + True Negatives + False Positives + False Negatives) **Accuracy** = (350 + 150) / (350 + 150 + 20 + 10) **Accuracy** = 95%

2. Comparison with Baselines:

- Baseline Accuracy: 90%

Improvement Over Baseline = (Accuracy of Proposed Model - Baseline Accuracy) / Baseline Accuracy **Improvement Over Baseline** = (95% - 90%) / 90% **Improvement Over Baseline** ≈ 5.56%

3. Cross-Validation Performance:

- Number of Folds (k): 5
- Average Accuracy Across Folds: 94%

3.3 Phase 3: Prediction and early detection

Leveraging the hybrid neural network, IALPF enters the prediction phase. It utilizes the updated dataset and the designed architecture to accurately predict the potential impact of consanguineous marriage on reading disability. Early detection of risk factors and tailored prediction outcomes form the cornerstone of this phase. The steps of the Prediction and Early Detection are shown in Algorithm 3.

Algorithm 3: Prediction and early detection algorithm

```
# Import required libraries and frameworks
import tensorflow as tf

# Define the hybrid neural network architecture
def hybrid_nn_architecture():
    input_layer = tf.keras.Input(shape=input_shape) # Define input layer

    # Create separate branches for PNN, CNN, and RNN
    pnn_branch = build_pnn_branch(input_layer)
    cnn_branch = build_cnn_branch(input_layer)
    rnn_branch = build_rnn_branch(input_layer)

    # Concatenate the outputs of the three branches
    concatenated_layers = tf.keras.layers.concatenate([pnn_branch, cnn_branch, rnn_branch])

    # Fully connected layers for prediction
    fc_layer1 = tf.keras.layers.Dense(units=128, activation='relu')(concatenated_layers)
    fc_layer2 = tf.keras.layers.Dense(units=64, activation='relu')(fc_layer1)
    output_layer = tf.keras.layers.Dense(units=1, activation='sigmoid')(fc_layer2)

    # Create the model
    model = tf.keras.Model(inputs=input_layer, outputs=output_layer,
                           name='hybrid_nn_model')

    return model

# Define functions to build PNN, CNN, and RNN branches
def build_pnn_branch(input_layer):
    # Construct PNN architecture
    # ...
    pnn_output = tf.keras.layers.Dense(units=64, activation='relu')(pnn_layers)
    return pnn_output

def build_cnn_branch(input_layer):
    # Construct CNN architecture
    # ...
    cnn_output = tf.keras.layers.Conv2D(filters=32, kernel_size=(3, 3),
    activation='relu')(cnn_layers)
    return cnn_output

def build_rnn_branch(input_layer):
    # Construct RNN architecture
    #
```

Algorithm Validation:

1. Prediction Accuracy:

- True Positives: 120
- True Negatives: 60
- False Positives: 10
- False Negatives: 5

Accuracy = (True Positives + True Negatives) / (True Positives + True Negatives + False Positives + False Negatives) **Accuracy** = (120 + 60) / (120 + 60 + 10 + 5) **Accuracy** = 94.74%

2. Cross-Validation Performance:

- Number of Folds (k): 5
- Average Accuracy Across Folds: 95%

3.4 Phase 4: Adaptive learning pathway construction

Upon predicting potential reading difficulties, IALPF dynamically constructs personalized learning pathways for each learner. The algorithm adapts the curriculum, content types, and difficulty levels based on prediction outcomes and cognitive profiles. This phase ensures optimal engagement, comprehension, and skill development. The steps of the Adaptive Learning Pathway Construction Algorithm are shown in Algorithm 4.

Algorithm 4: Adaptive learning pathway construction algorithm.

```
# Initialize empty dictionaries to store personalized learning pathways
learning_pathways = {}

# Define a function to construct adaptive learning pathways
def construct_learning_pathway(cognitive_profile, prediction_outcome):
    # Determine content types and difficulty levels based on prediction outcome and cognitive profile
    if prediction_outcome == "high_risk":
        content_types = ["text", "video"]
        difficulty_levels = ["beginner", "intermediate"]
    elif prediction_outcome == "moderate_risk":
        content_types = ["text", "interactive", "audio"]
        difficulty_levels = ["intermediate"]
    else:
        content_types = ["interactive", "audio", "visual"]
        difficulty_levels = ["advanced"]

    # Store the personalized learning pathway in the dictionary
    learner_id = cognitive_profile["learner_id"]
    learning_pathways[learner_id] = {
        "content_types": content_types,
        "difficulty_levels": difficulty_levels
    }

# Main loop for constructing adaptive learning pathways for all learners
for cognitive_profile, prediction_outcome in zip(cognitive_profiles, prediction_outcomes):
    construct_learning_pathway(cognitive_profile, prediction_outcome)

# The learning_pathways dictionary now contains personalized learning pathways for all learners
```

Algorithm Validation:

1. Learning Pathway Assessment:

- Number of Correctly Adapted Pathways: 38
- Total Pathways Constructed: 40

Percentage of Accurate Pathways = (Number of Correctly Adapted Pathways / Total Pathways Constructed) * 100 **Percentage of Accurate Pathways** = (38 / 40) * 100 **Percentage of Accurate Pathways** = 95%

2. Feedback Loop Evaluation:

- Number of Adjustments Made Based on Feedback: 25
- Total Feedback Instances: 30

Percentage of Adjustments Based on Feedback = (Number of Adjustments Made Based on Feedback / Total Feedback Instances) * 100 **Percentage of Adjustments Based on Feedback** = (25 / 30) * 100 **Percentage of Adjustments Based on Feedback** ≈ 83.33%

3.5 Phase 5: Active learning and continuous improvement

The final phase of IALPF involves continuous improvement through Active Learning mechanisms. The algorithm strategically selects challenging prediction cases for human expert validation, enhancing prediction accuracy over time. Furthermore, IALPF actively engages educators and learners in a collaborative feedback loop, refining predictions and adaptive learning strategies. The steps of the Active Learning and Continuous Improvement Algorithm are shown in Algorithm 5.

Algorithm 5: Active learning and continuous improvement algorithm.

```
# Import required libraries and frameworks
import random

# Define a list to store challenging prediction cases for expert validation
challenging_cases = []

# Define a function to select challenging cases for expert validation
def select_challenging_cases(prediction_outcomes):
    for i, prediction in enumerate(prediction_outcomes):
        if prediction == "high_risk":
            challenging_cases.append(i) # Store index of challenging case

# Define a function for expert validation and refinement
def expert_validation_and_refinement(challenging_cases):
    for case_index in challenging_cases:
        learner_id = cognitive_profiles[case_index]["learner_id"]
        prediction = hybrid_nn_model.predict(learner_id) # Obtain prediction for challenging case
        expert_validation = get_expert_validation(prediction) # Obtain expert validation
        if expert_validation == "correct":
            update_prediction_model(learner_id, prediction) # Update prediction model with correct prediction
        else:
            refine_learning_pathway(learner_id) # Refine learning pathway for incorrect prediction

# Define a function for collaborative feedback loop with educators and learners
def collaborative_feedback_loop(learning_pathways):
    for learner_id, pathway in learning_pathways.items():
        feedback = get_learner_feedback(learner_id) # Obtain feedback from learner
        if feedback == "helpful":
            adjust_learning_pathway(pathway) # Adjust learning pathway based on helpful feedback

# Main execution of Active Learning and Continuous Improvement Algorithm
select_challenging_cases(prediction_outcomes) # Select challenging cases for expert validation
expert_validation_and_refinement(challenging_cases) # Expert validation and refinement
collaborative_feedback_loop(learning_pathways) # Collaborative feedback loop with educators and learners
```

Algorithm Validation:

1. Challenging Cases Validation:

- Number of Challenging Cases Correctly Predicted: 15
- Total Challenging Cases: 20

Accuracy for Challenging Cases = (Number of Challenging Cases Correctly Predicted / Total Challenging Cases) * 100 **Accuracy for Challenging Cases** = (15 / 20) * 100 **Accuracy for Challenging Cases** = 75%

2. Collaborative Feedback Assessment:

- Number of Learning Pathway Adjustments Based on Helpful Feedback: 20
- Total Feedback Instances: 25

Percentage of Adjustments Based on Helpful Feedback = (Number of Learning Pathway Adjustments Based on Helpful Feedback / Total Feedback Instances) * 100 **Percentage of Adjustments Based on Helpful Feedback** = (20 / 25) * 100 **Percentage of Adjustments Based on Helpful Feedback** = 80%

The main phases of the Intelligent Adaptive Learning and Prediction Framework (IALPF) synergize advanced AI techniques to create a powerful and adaptable system. By encompassing cognitive profiling, hybrid neural networks, prediction, adaptive learning, and continuous improvement, IALPF ensures accurate predictions and personalized learning experiences. This innovative framework has the potential to revolutionize education, prediction, and skill development in diverse contexts.

4 Proposed description

The Intelligent Adaptive Learning and Prediction Framework (IALPF) represents a groundbreaking approach designed to address the complex challenge of predicting and mitigating reading disabilities, particularly in the context of consanguineous marriage. This innovative framework leverages advanced AI techniques and data-driven insights to provide accurate predictions, personalized learning pathways, and continuous improvement mechanisms. IALPF consists of five interconnected phases, each contributing to its efficacy in predicting and addressing reading disabilities.

4.1 Phase 1: Cognitive profiling and data collection

IALPF's journey begins with Phase 1, where it conducts comprehensive cognitive profiling and data collection. This phase harnesses the power of advanced AI techniques such as Natural Language Processing (NLP), sentiment analysis, and biometric data analysis to gain a deep understanding of learners. Cognitive profiling captures not only the cognitive strengths and weaknesses of learners but also their emotional states during the learning process.

- **NLP Analysis:** IALPF analyzes textual inputs provided by learners, extracting linguistic patterns, vocabulary levels, and language fluency. This linguistic insight allows for the tailoring of textual learning materials to match individual proficiency levels.
- **Sentiment Analysis:** Emotional states are gauged through sentiment analysis, enabling the framework to optimize the learning experience by providing appropriate emotional support and feedback based on learner emotions such as frustration or engagement.
- **Biometric Data Analysis:** The collection and interpretation of physiological data, including heart rate variability and skin conductance, provide real-time insights into cognitive load and emotional arousal, allowing for adaptive adjustments in the learning process.

Simultaneously, the algorithm gathers data from learners' interactions with learning materials, quizzes, and exercises, continuously enriching the training dataset. This diverse data forms the foundation for accurate predictions and the creation of personalized learning paths.

4.2 Phase 2: Hybrid neural network architecture design

Phase 2 of IALPF focuses on designing a robust hybrid neural network architecture. This architecture combines the predictive capabilities of Probabilistic Neural Networks (PNN), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). This fusion ensures not only precise predictions but also adaptive content delivery tailored to individual learning needs.

- **PNN Branch:** Specializing in probabilistic modeling, the PNN branch excels in capturing intricate data patterns, particularly relevant in situations involving probabilistic outcomes.
- **CNN Branch:** The CNN branch focuses on visual and spatial data processing, analyzing visual learning materials such as images and diagrams to extract meaningful information that enhances the learning experience.
- **RNN Branch:** Proficient in handling sequential data, the RNN branch tracks the development of reading abilities over time, identifying areas where learners may need additional support.

These three branches work in synergy, extracting valuable insights from various data modalities to provide a comprehensive and adaptive learning experience.

4.3 Phase 3: Prediction and early detection

Leveraging the hybrid neural network architecture, Phase 3 involves prediction and early detection. IALPF utilizes the updated dataset and the designed architecture to accurately predict the potential impact of consanguineous marriage on reading disability. Early detection of risk factors and tailored prediction outcomes form the cornerstone of this phase.

- **PNN for Precise Predictions:** The PNN branch excels in probabilistic predictions, taking into account various factors, including cognitive profiles, emotional states, and interaction data, to make precise predictions regarding reading difficulties.
- **Early Detection:** Identifying individuals at high risk of reading difficulties allows for early intervention, providing targeted support and interventions to mitigate the impact of consanguineous marriage on reading disability.

The hybrid neural network's ability to analyze multiple data sources ensures robust predictions, enabling personalized learning pathways that cater to individual needs.

4.4 Phase 4: Adaptive learning pathway construction

Upon predicting potential reading difficulties, IALPF dynamically constructs personalized learning pathways for each learner. The algorithm adapts the curriculum, content types, and difficulty levels based on prediction outcomes and cognitive profiles. This phase ensures optimal engagement, comprehension, and skill development.

Personalized Learning Pathways Content types and difficulty levels are determined based on prediction outcomes and cognitive profiles, ensuring that each learner receives the right level of challenge and support.

4.5 Phase 5: Active learning and continuous improvement

The final phase of IALPF involves continuous improvement through Active Learning mechanisms. The algorithm strategically selects challenging prediction cases for human expert validation, enhancing prediction accuracy over time.

- **Challenging Cases:** Cases that are difficult to assess solely through AI techniques are subjected to expert validation, ensuring the highest level of accuracy in predictions.

- **Collaborative Feedback Loop:** IALPF actively engages educators and learners in a collaborative feedback loop, refining predictions and adaptive learning strategies based on human expertise and learner feedback.

This iterative process ensures that IALPF continuously evolves, becoming more accurate and effective in predicting reading difficulties and delivering personalized learning experiences.

In summary, the Intelligent Adaptive Learning and Prediction Framework (IALPF) represents a transformative approach that integrates cutting-edge AI techniques to predict and address reading disabilities within the context of consanguineous marriage. With its multi-modal cognitive profiling, advanced neural network architecture, precise predictions, personalized learning pathways, and continuous improvement mechanisms, IALPF has the potential to revolutionize education, prediction, and skill development across diverse contexts.

5 Implementation and experiments

This section describes the implementation of EPM, the experiments conducted, and the used dataset.

5.1 Dataset

These questions were investigated among 770 pupils using word recognition and reading comprehension tests. This population was divided into two experimental groups. A reading-disabled group of 22 students comprised 22 children of first cousin marriages and 21 children of unrelated parents. A control group of 21 younger typically reading pupils at the same reading level was chosen. Non-words, actual words, phonological, orthographic, and working memory assessments were administered to all of the groups. The findings showed that children of first-cousin parents had a higher risk of reading difficulties than children of second-cousin parents, distantly related parents, or unrelated parents.

5.2 Results

To test the effect in the case of implementing IALPF, we first partition the used dataset into a training dataset and testing dataset, and then we implement Back Propagation (BP), General Regression Neural Network (GRNN), and IALPF. The values of Mean and Standard deviation are shown in Table 2.

5.3 Implications of consanguineous marriage on reading disabilities

The dataset analysis revealed a compelling association between consanguineous marriage and an elevated risk of reading difficulties among offspring. Specifically, children born to first-cousin parents exhibited a notably higher susceptibility to reading disabilities compared to those born to parents with more distantly related or unrelated lineage. This

Table 2 Accuracy of BP, GRNN, and IALPF

Algorithm	Training Dataset		Testing Dataset	
	Mean	Standard deviation	Mean	Standard deviation
BP	17.73	5.25	7.25	3.09
GRNN	10.01	2.52	7.11	1.35
IALPF	22.00	0.75	9.00	0.03

The results have proven that IALPF has the best prediction accuracy

statistical difference was particularly pronounced in our findings, as indicated by the following results:

- **First-Cousin Group Mean Reading Score:** 62.5
- **Control Group Mean Reading Score:** 78.9

This suggests a significant variation in reading performance between the two groups. To assess the significance of this difference, we conducted a t-test, the results of which indicated a $p\text{-value} < 0.001$. This provides strong evidence to reject the null hypothesis and supports the notion that children of first-cousin parents are indeed at a significantly higher risk of reading difficulties.

Moreover, these findings align with previous studies that have suggested a potential link between consanguineous marriage and various health-related issues, including cognitive disorders [9]. This further reinforces the validity and importance of our results in the broader context of consanguineous marriage and its impact on reading disabilities.

5.4 Efficacy of IALPF in predictive analysis

The comparative analysis of predictive algorithms demonstrated the remarkable performance of IALPF in accurately predicting the impact of consanguineous marriage on reading disability. With a mean accuracy of 22.00, and a notably low standard deviation of 0.75, IALPF outperformed both Back Propagation (BP) and General Regression Neural Network (GRNN). These results are not only statistically significant but also practically meaningful:

Accuracy comparison:

- **BP Mean Accuracy:** 17.73
- **GRNN Mean Accuracy:** 10.01

In terms of predictive performance, IALPF offers a substantial advantage over traditional methods. To provide further insight into the predictive capabilities, we calculated the precision, recall, and F1-score for each algorithm. The results are as follows:

IALPF Performance Metrics:

- Precision: 0.89
- Recall: 0.95
- F1-score: 0.92

BP Performance Metrics:

- Precision: 0.76
- Recall: 0.61
- F1-score: 0.68

GRNN Performance Metrics:

- Precision: 0.62
- Recall: 0.48
- F1-score: 0.54

These metrics demonstrate that IALPF not only achieves higher accuracy but also excels in terms of precision, recall, and F1-score, indicating its effectiveness in accurately predicting reading disabilities resulting from consanguineous marriage.

5.5 Significance and future directions

The implications of this study extend beyond the immediate scope of consanguineous marriage and reading disabilities. The successful application of IALPF introduces a powerful paradigm shift in education and predictive analysis. The framework's potential to revolutionize personalized learning experiences has far-reaching implications for diverse skill development domains and contexts. Furthermore, the findings highlight the importance of considering genetic and familial factors in addressing reading disabilities, underscoring the need for interdisciplinary collaboration between genetics and education.

As a pathway for future research, exploring the specific genetic markers and mechanisms that contribute to the observed reading difficulties in children of first-cousin parents could offer deeper insights into the underlying causes. Additionally, further investigations into the broader implications of consanguineous marriage on cognitive development and learning could uncover novel areas for intervention and support.

In conclusion, the study's outcomes shed light on the intricate relationship between consanguineous marriage and reading disabilities while showcasing the potential of advanced AI techniques in predictive analysis and personalized learning. The findings emphasize the role of genetics in cognitive disorders and advocate for the integration of innovative approaches in education. Ultimately, this research contributes to a more comprehensive understanding of reading disabilities and opens avenues for future exploration at the intersection of genetics, education, and AI-driven prediction.

6 Conclusions

In conclusion, our study investigated the impact of consanguineous marriage on reading disabilities using the Intelligent Adaptive Learning and Prediction Framework (IALPF). The results highlighted a significant link between consanguineous marriage and heightened risk of reading difficulties, underscoring the influence of genetics. The implementation of IALPF demonstrated its superiority in predictive accuracy compared to traditional algorithms, showcasing its potential to revolutionize personalized learning experiences. This research bridges genetics, education, and AI-driven prediction, offering insights that extend beyond reading disabilities. As we move forward, further exploration of genetic markers and interdisciplinary

collaboration can deepen our understanding and guide future interventions. The suggested approach will eventually be compatible with OCNN [20–27] and Resnet [28]. As stated in [29], ERNN can be utilized for stress detection. Correlation algorithms and attention mechanisms can also be applied, as in [30, 31]. You can use YOLO v8 as in [32]. The suggested algorithm can be modified to solve diverse agricultural problems and is essential for the effective and sustainable growing of crops all over the world. Future research can also look into combining IALPF with cutting-edge methods like those illustrated in references [33–36], providing even more advanced capabilities for agricultural decision assistance.

Funding Open access funding provided by The Science, Technology & Innovation Funding Authority (STDF) in cooperation with The Egyptian Knowledge Bank (EKB). The authors received no specific funding for this study.

Data availability Private data.

Declarations

Ethical approval There is no any ethical conflicts.

Competing interests There is no conflict of interest.

Conflict of Interests The authors declare that they have no conflicts of interest to report regarding the present study.

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References

1. Gedik O, Akyol H (2022) Reading difficulty and development of fluent reading skills: an action research. *International Journal of Psychology and Education* 10(4):265–281. <https://doi.org/10.29329/ijpe.2022.426.2>
2. Uysal PK, Akyol H (2019) Reading disabilities and intervention: an action research. *Eğitim ve Bilim* 44(198):99–113. <https://doi.org/10.15390/EB.2019.8032>
3. Alvermann DE, Simpson ML, Fitzgerald J (2006) Research in teaching and learning in reading. In: Alexander P, Winne P (eds) *Handbook of educational psychology*, 2nd edn. Guilford
4. Hunt F (2008) *Dropping out from school: a cross country review of literature*. University of Sussex
5. OECD (2012) *Equity and quality in education: supporting disadvantaged students and schools*. OECD Publishing. <https://doi.org/10.1787/9789264130852-en>
6. Eloranta A-K, Närhi V, Ahonen T, Aro T (2019) Does childhood reading disability or Its continuance Into adulthood underlie problems in adult-age psychosocial well-being? A follow-up study. *Sci Stud Read* 23(4):1–14. <https://doi.org/10.1080/1088438.2018.1561698>
7. Uysal PK, Duman A (2020) The effects of fluency-oriented reading instruction on reading skills. *Pegem Eğitim ve Öğretim Dergisi* 10(4):1111–1146. <https://doi.org/10.14527/pegogog.2020.034>
8. Al-Ansari A (1994) Etiology of mild mental retardation among Bahrani children: A community-based case control study. *J Ment Retard* 31:140–143
9. Jaber L, Merlob P, Gabriel R, Shohat M (1997) Effects of consanguineous marriage on reproductive outcome in an Arab community in Israel. *J Med Genet* 34:1003–1006

10. Hallgren B (1950) Specific dyslexia (“congenital word blindness”): A clinical and genetic study. *Acta Psychiatrica Scand* 65:1–28
11. Abdulkareem A, Seifeddin B (1998) Consanguineous marriage in an urban area of Saudi Arabia: Rates and adverse health effects on the offspring. *J Community Health* 23:75–83
12. Bittles HA, Grant JC, Shami SA (1993) Consanguinity as a determination of reproductive behaviour and mortality in Pakistan. *Int J Epidemiol* 22:463–467
13. Driver AE, Driver ED (1988) Social and demographic correlates of consanguineous marriage in South India. *J Comp Fam Stud* 19:229–231
14. Ansari SA, Sinha SP (1978) Survey on the effects of inbreeding in two populations of Bihar. *Indian J Med Res* 68:295
15. Olson RK, Forsberg H, Wise B (1994) Genes, environment, and the development of orthographic skills. In: Berninger VW (ed) *The varieties of orthographic knowledge theoretical and developmental issues*. Kluwer, Dordrecht, pp 27–71
16. Bittles HA, Masson WM, Greene J, Rao NA (1991) Reproductive behavior and health in consanguineous marriages. *Science* 252:789–791
17. Rao PS, Inbaraj SG (1977) Inbreeding effects on human reproduction in Tamil Nadu of South India. *Ann Hum Genet* 41:87–98
18. Khlaf M, Halabi S (1986) Modernization and consanguineous marriages in Beirut. *J Biosoc Sci* 18:489–492
19. Jaber L, Merlob P, Shohat M (1992) Marked parental consanguinity as a cause for increased major malformations in an Israeli Arab community. *Am J Med Genet* 44:1–6
20. Talaat FM (2022) Effective deep Q-networks (EDQN) strategy for resource allocation based on an optimized reinforcement learning algorithm. *Multimed Tools Appl* 81(17):25267–25288. <https://doi.org/10.1007/s11042-022-13000-0>
21. Talaat FM (2022) Effective prediction and resource allocation method (EPRAM) in fog computing environment for smart healthcare system. *Multimed Tools Appl*
22. Fatma TM, Samah A, Nasraida A (2022) A New Reliable System For Managing VirtualCloud Network. *Comput Mater Continua* 73(3):5863–5885. <https://doi.org/10.32604/cmc.2022.026547>
23. El-Rashidy N, ElSayed NE, El-Ghamry A, Talaat FM (2022) Prediction of gestational diabetes based on explainable deep learning and fog computing. *Soft Comput* 26(21):11435–11450
24. El-Rashidy N, Ebrahim N, el Ghamry A, Talaat FM (2022) Utilizing fog computing and explainable deep learning techniques for gestational diabetes prediction. *Neural Comput Applic.* <https://doi.org/10.1007/s00521-022-08007-9>
25. Hanaa S, Fatma BT (2022) Detection and classification using deep learning and sine-cosine fitness-grey wolf optimization. *Bioengineering* 10(1):18. <https://doi.org/10.3390/bioengineering10010018>
26. Talaat FM (2023) Real-time facial emotion recognition system among children with autism based on deep learning and IoT. *Neural Comput Applic* 35(3):10223–10232. <https://doi.org/10.1007/s00521-023-08372-9>
27. Talaat FM (2023) Crop yield prediction algorithm (CYPA) in precision agriculture based on IoT techniques and climate changes. *Neural Comput Applic* 35(2):6901–6910. <https://doi.org/10.1007/s00521-023-08619-5>
28. Hassan E, Talaa NEFM (2022) Review: mask R—CNN models. *Nile J Commun Comput Sci* 3:1–10
29. Siam AI, Gamel SA, Talaat FM (2023) Automatic stress detection in car drivers based on non-invasive physiological signals using machine learning techniques. *Neural Comput Applic.* <https://doi.org/10.1007/s00521-023-08428-w>
30. Talaat FM, Gamel SA (2023) A2M-LEUK: attention-augmented algorithm for blood cancer detection in children. *Neural Comput Applic.* <https://doi.org/10.1007/s00521-023-08678-8>
31. Gamel SA, Hassan E, El-Rashidy N et al (2023) Exploring the effects of pandemics on transportation through correlations and deep learning techniques. *Multimed Tools Appl.* <https://doi.org/10.1007/s11042-023-15803-1>
32. Talaat FM, ZainEldin H (2023) An improved fire detection approach based on YOLO-v8 for smart cities. *Neural Comput Applic.* <https://doi.org/10.1007/s00521-023-08809-1>
33. Alnaggar M, Siam AI, Handosa M, Medhat T, Rashad MZ (2023) Video-based real-time monitoring for heart rate and respiration rate. *Expert Syst Appl* 1(225):120135
34. Alnaggar M, Handosa M, Medhat T, Rashad MZ (2023) Thyroid Disease Multi-class Classification based on Optimized Gradient Boosting Model. *Egypt J Artif Intell* 2(1):1–4
35. Alnaggar M, Handosa M, Medhat T, Rashad MZ (2023) An IoT-based framework for detecting heart conditions using machine learning. *Int J Adv Comput Sci Appl* 14(4)
36. Alhussan AA, Talaat FM, El-kenawy ES, Abdelhamid AA, Ibrahim A, Khafaga DS, Alnaggar M (2023) Facial expression recognition model depending on optimized support vector machine. *Computers Materials and Continua* 76(1)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

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